1 The Group 17 elements are oxidising agents.

(a) **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX.

$$3X_2 + 2NH_3 \rightarrow N_2 + 6HX \qquad X = Cl \text{ or Br}$$

The relative bond strengths of X–X and H–X determine the difference in enthalpy change of the two reactions.

(i) Describe and explain the difference in the X–X bond strengths of Cl_2 and Br_2.

.....

.....

.....

..... [2]

(iii) Define enthalpy change of formation, ΔH_f.

.....

.....

..... [2]

(iv) Table 1.1 gives data relevant to the reaction of $Cl_2(g)$ with $NH_3(g)$.

Table 1.1

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
$NH_3(g)$	–46
$HCl(g)$	–92

Use the data in Table 1.1 to calculate the enthalpy change of the reaction of $Cl_2(g)$ with $NH_3(g)$.

enthalpy change of reaction = ... kJ mol^{-1} [2]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

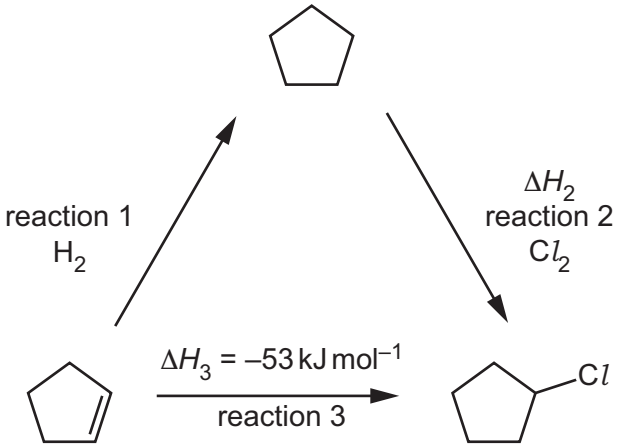


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(ii) Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2.

Table 3.1

compound	enthalpy change of combustion, ΔH_c / kJ mol^{-1}
	–3292
	–3115
H_2	–286

ΔH_2 = ... kJ mol^{-1} [2]

3 The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

.....

..... [2]

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

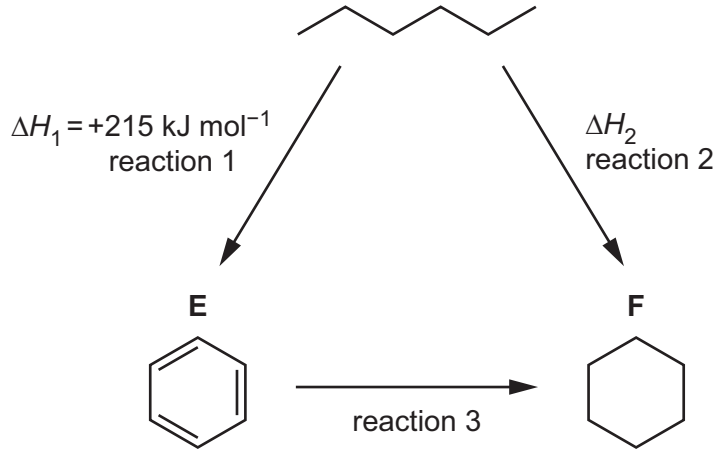


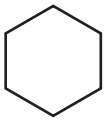
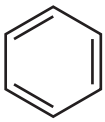
Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(ii) Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2.

Table 3.1

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
	−156
	+48

ΔH_2 = ... kJ mol^{-1} [2]

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

.....

..... [1]

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl.

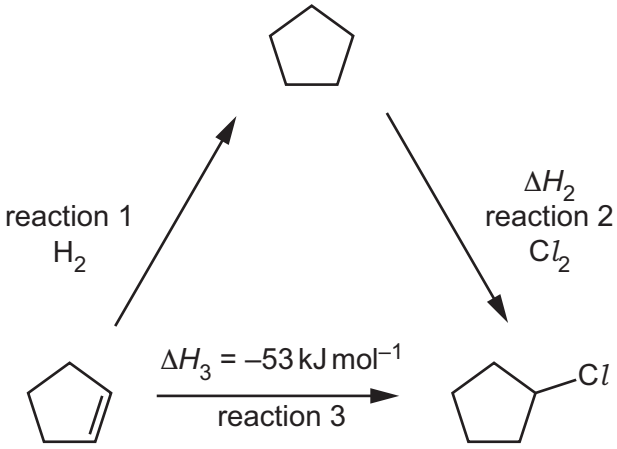


Fig. 3.2

(i) Identify a suitable reagent for reaction 3.

..... [1]

(ii) Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2.

Table 3.1

compound	enthalpy change of combustion, ΔH_c / kJ mol^{-1}
	–3292
	–3115
H_2	–286

ΔH_2 = ... kJ mol^{-1} [2]

3 **(a)** C_3H_6 reacts with HBr(g) in an addition reaction.

(i) Define addition reaction.

.....

..... [1]

(c) Hydrocarbon **Y** contains two C=C groups. There are no other functional groups present.

Y reacts with an excess of H_2(g) to produce hexane, C_6H_{14}.

Table 3.1

bond	energy / kJ mol^{-1}
C−C	350
C=C	610
C≡C	840
H−H	436
C−H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = ... kJ mol^{-1} [2]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

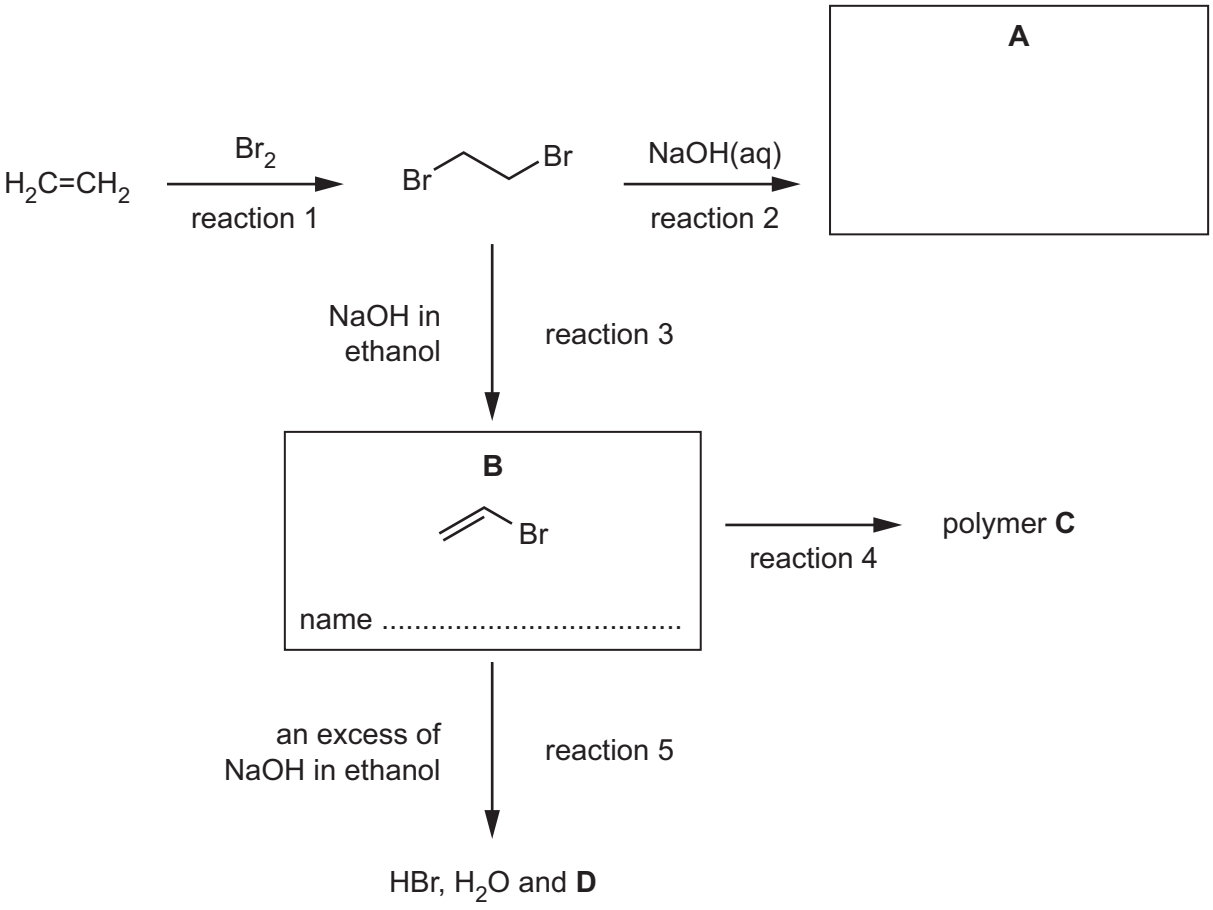


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

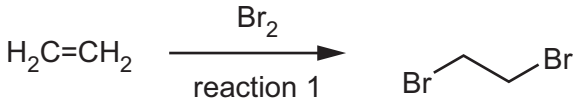
Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(b) The enthalpy change of reaction 1, $\Delta H_r = -90.0\,kJ\,mol^{-1}$.



The enthalpy change of formation of ethene, $\Delta H_f = +52.2\,kJ\,mol^{-1}$.

Calculate the enthalpy change of formation of 1,2-dibromoethane.

ΔH_f of 1,2-dibromoethane = ... $kJ\,mol^{-1}$ [1]

3 **(a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O. Include state symbols.

..... [2]

(b) Water is one of the products in the reaction of B_2O_3 and NH_3, as shown in reaction 2.

reaction 2 $B_2O_3 + 2NH_3 \rightarrow 2BN + 3H_2O$

Table 3.1 shows information about the standard enthalpy change of formation, $\Delta H_f^{\ominus}$, of some substances.

Table 3.1

substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
B_2O_3	–1264
NH_3	–46
BN	–134
H_2O	–286

Calculate the enthalpy change, ΔH, for reaction 2 using the data from Table 3.1.

ΔH = ... kJ mol^{-1} [2]

4 **(a)** $NH_3(g)$ reacts with $HCl(g)$ to produce $NH_4Cl(s)$, as shown.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl.

[2]

(d) When 2-bromo-2-methylpropane reacts with OH^-, two mechanisms, S_N1 and S_N2, both occur. The S_N2 mechanism has a slower rate.

Fig. 4.1 shows the reaction pathway diagram for the S_N1 mechanism.

Sketch a graph on Fig. 4.1 to show the reaction pathway for the S_N2 mechanism.

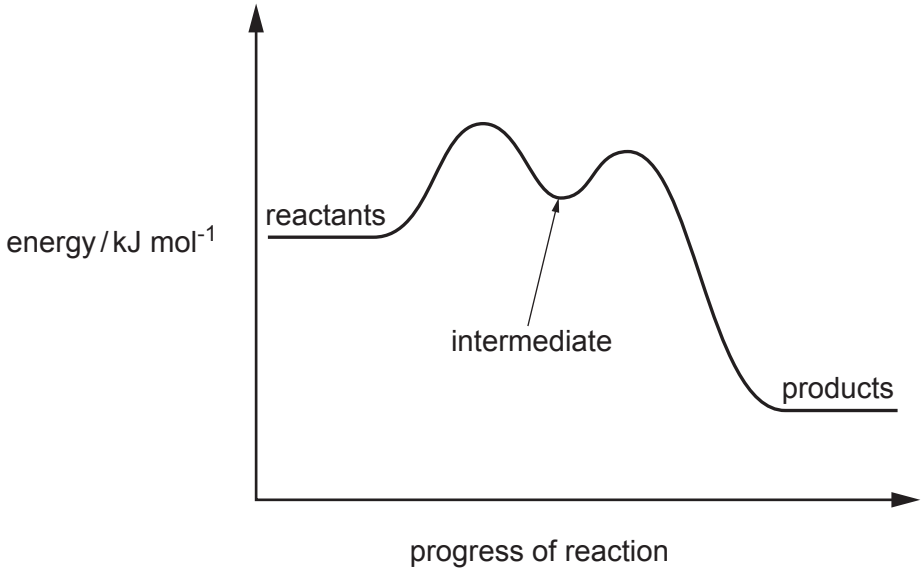


Fig. 4.1

[2]

2 Separate samples of Na_2CO_3 and $NaHCO_3$ react with HC*l*(aq) to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	ΔH / kJ mol^{-1}
1	$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$	ΔH_1
2	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	$\Delta H_2 = +27.2$

(a) Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2, and the activation energy, E_A.

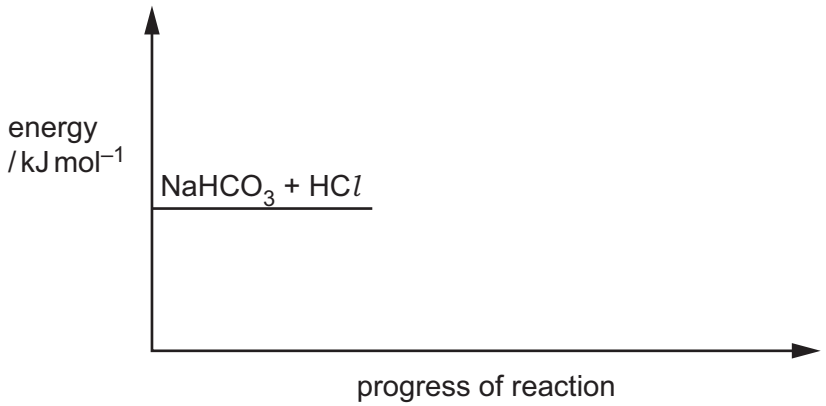


Fig. 2.1

[2]

(b) The value for ΔH_1 is determined by experiment using the following method.

- 50.0 cm^3 of 2.00 mol dm^{-3} HC*l*(aq) is added to a polystyrene cup.
- The initial temperature of the acid is recorded as 19.6 °C.
- 0.0400 mol of Na_2CO_3 is added and the mixture is stirred.
- All the solid Na_2CO_3 disappears and a colourless solution is produced.

The maximum temperature recorded during the reaction is 26.2 °C.

(i) Describe **one** other observation that shows the reaction is complete.

..... [1]

(ii) Calculate the value of ΔH_1 in $kJ\,mol^{-1}$.

Assume the specific heat capacity of the reaction mixture is the same as for water and no heat is lost to the surroundings.

Show your working.

ΔH_1 = ... $kJ\,mol^{-1}$ [3]

(iii) Thermal decomposition occurs when $NaHCO_3$ is heated.

reaction 3 $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$

Calculate the enthalpy change for reaction 3, ΔH_r, using the data in Table 2.1 and the value of ΔH_1 calculated in **(b)(ii)**.

(If you were unable to calculate a value for ΔH_1 in **(b)(ii)**, assume the enthalpy change is $-38.4\,kJ\,mol^{-1}$. This is **not** the correct value.)

ΔH_r = ... $kJ\,mol^{-1}$ [2]

4 **(a)** Define enthalpy change of formation.

.....

.....

..... [2]

(b) Iron is made when iron(III) oxide is heated with carbon monoxide, as shown by reaction 2.

reaction 2 $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Table 4.1 shows enthalpy change of formation data measured at 298 K and 101 kPa.

Table 4.1

substance	equation	value for $\Delta H_f^{\ominus}$ / kJ mol^{-1}
Fe_2O_3		–824.2
CO		–110.5
CO_2	$C(s) + O_2(g) \rightarrow CO_2(g)$	–393.5

(i) Complete Table 4.1 by adding equations with relevant state symbols to represent:

- standard enthalpy change of formation for Fe_2O_3
- standard enthalpy change of formation for CO.

[2]

(ii) Use the data in Table 4.1 to calculate the enthalpy change of reaction, ΔH_r, in kJ mol^{-1}, for reaction 2.

Show your working.

ΔH_r = ... kJ mol^{-1} [2]

3 Nitrogen, N_2, is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1 .....

2 .....

[2]

(c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3.

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H = -92\,kJ\,mol^{-1}$$

(i) Use the information in **(c)** to complete Table 3.1.

Table 3.1

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
N_2	
H_2	
NH_3	

[2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(e) $POCl_3(g)$ forms when $PCl_3(g)$ reacts with $O_2(g)$.

$$2PCl_3(g) + O_2(g) \longrightarrow 2POCl_3(g)$$

Table 1.3 gives some relevant data.

Table 1.3

process	value / kJ mol^{-1}
enthalpy change of formation of $PCl_3(g)$	–289
enthalpy change of formation of $POCl_3(g)$	–592
$O_2(g) \longrightarrow 2O(g)$	+496

(i) Define enthalpy change of formation, ΔH_f.

.....

.....

..... [2]

(ii) Calculate the bond energy of P=O in $POCl_3$ using the data in Table 1.3.

Show your working.

bond energy of P=O = ... kJ mol^{-1}

[2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(b) $KClO_3(s)$ decomposes when heated.

$MnO_2(s)$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $MnO_2(s)$ on the decomposition of $KClO_3(s)$.

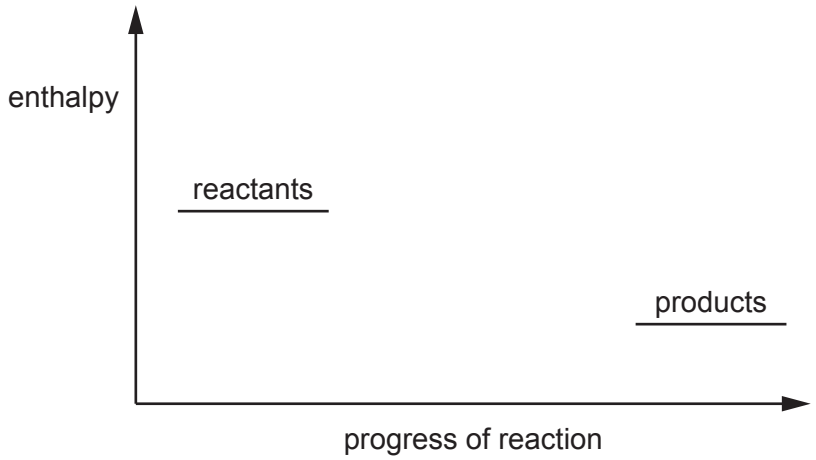


Fig. 3.1

[2]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(b) H_3PO_4 is also formed in the process shown in reaction 1.

reaction 1 $4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$

Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
H_3PO_3	–972
H_3PO_4	–1281
PH_3	+9

(i) Define enthalpy change of formation.

.....

.....

..... [2]

(ii) Use the data in Table 3.1 to calculate the enthalpy change, ΔH_r, of reaction 1.

ΔH_r = ... kJ mol^{-1}

[2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) **(i)** Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(e) $POCl_3(g)$ forms when $PCl_3(g)$ reacts with $O_2(g)$.

$$2PCl_3(g) + O_2(g) \longrightarrow 2POCl_3(g)$$

Table 1.3 gives some relevant data.

Table 1.3

process	value / kJ mol^{-1}
enthalpy change of formation of $PCl_3(g)$	–289
enthalpy change of formation of $POCl_3(g)$	–592
$O_2(g) \longrightarrow 2O(g)$	+496

(i) Define enthalpy change of formation, ΔH_f.

.....

.....

..... [2]

(ii) Calculate the bond energy of P=O in $POCl_3$ using the data in Table 1.3.

Show your working.

bond energy of P=O = ... kJ mol^{-1}

[2]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(b) $KClO_3(s)$ decomposes when heated.

$MnO_2(s)$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $MnO_2(s)$ on the decomposition of $KClO_3(s)$.

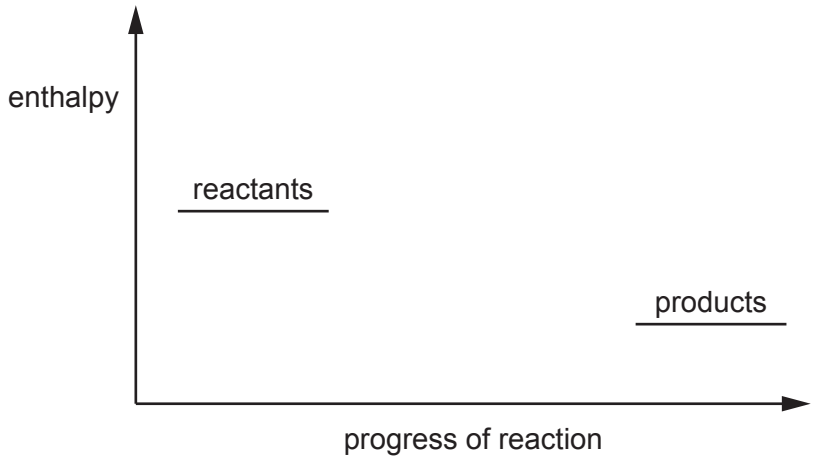


Fig. 3.1

[2]

3 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(b) In an experiment, 50.0 cm^3 of 2.00 mol dm^{-3} NaOH(aq) is added to 60.0 cm^3 of 1.00 mol dm^{-3} H_2SO_4(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

(i) Use equation 1 to calculate the amount, in mol, of H_2SO_4(aq) that is neutralised in the experiment.

amount of H_2SO_4 neutralised = ... mol [1]

(ii) Calculate the theoretical maximum temperature of the mixture in this experiment.

Assume that:

- enthalpy change of neutralisation, ΔH_{neut}, of NaOH(aq) and H_2SO_4(aq) is –57.1 kJ mol^{-1}
- full dissociation of H_2SO_4(aq) occurs
- the specific heat capacity of the final solution is 4.18 J g^{-1} K^{-1}
- 1.00 cm^3 of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- the experiment takes place at constant pressure.

Show your working.

theoretical maximum temperature = ... °C [3]

(c) The enthalpy change of neutralisation of $CH_3COOH(aq)$ and $NaOH(aq)$ is $-55.2\,kJ\,mol^{-1}$.

(i) Complete the equation for the reaction.

.....CH_3COOH + ...NaOH → ..... [1]

(ii) Values for the enthalpy change of neutralisation, ΔH_{neut}, are shown in Table 3.1.

Table 3.1

reagents	ΔH_{neut} / kJ mol^{-1}
NaOH + HC*l*	–57.1
NaOH + CH_3COOH	–55.2

Suggest why the value for ΔH_{neut} of the weak acid, CH_3COOH, reacting with NaOH is different to the value obtained using the strong acid, HC*l*. Assume that the values are determined under the same conditions.

.....

..... [1]

4 **(a)** Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

..... [1]

(c) The experiment is repeated at 500 K. The value of K_c under these conditions is 0.00625.

(i) Describe the difference in the composition of the equilibrium mixture at 500 K compared to 764 K.

.....

..... [1]

(ii) Use Le Chatelier's principle to deduce whether the decomposition of HI(g) is endothermic or exothermic. Explain your answer.

.....

.....

..... [1]

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4.

reaction 1 $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$

(a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r, for reaction 1.

Table 4.1

bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

ΔH_r = ... kJ mol^{-1} [2]

2 A neutralisation reaction occurs when NaOH(aq) is added to H_2SO_4(aq).

equation 1 $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$

(a) Define enthalpy change of neutralisation, ΔH_{neut}.

.....

.....

..... [2]

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between NaOH(aq) and H_2SO_4(aq).

100 cm^3 of 1.00 mol dm^{-3} NaOH(aq) is added to 75 cm^3 of 1.00 mol dm^{-3} H_2SO_4(aq) in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of NaOH(aq) / °C	20.0
initial temperature of H_2SO_4(aq) / °C	20.0
maximum temperature of mixture / °C	27.8

(i) Use equation 1 to calculate the amount, in mol, of H_2SO_4(aq) that is neutralised in the experiment.

amount of H_2SO_4(aq) neutralised = ... mol [1]

(ii) Calculate ΔH_{neut} using the results in Table 2.1. Include units in your answer.

Assume that:
- the specific heat capacity of the final solution is 4.18 J g^{-1} K^{-1}
- 1.00 cm^3 of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- full dissociation of H_2SO_4(aq) occurs
- the experiment takes place at constant pressure.

Show your working.

ΔH_{neut} = ... units .....
[3]

(c) **(i)** Complete the equation for the reaction that occurs when a solution of $Ba(OH)_2$ is added to aqueous sulfuric acid. Include state symbols.

..... $H_2SO_4(aq)$ + ...$Ba(OH)_2(aq)$ → ……

[2]

(ii) Suggest why the enthalpy change of neutralisation cannot be determined using the addition of dilute sulfuric acid to aqueous barium hydroxide.

.....

..... [1]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

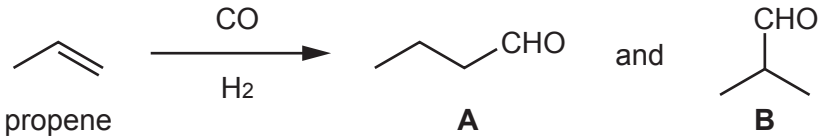


Fig. 3.1

(a) **(i)** Define addition reaction.

.....

..... [1]

(d) The reaction of ethene, C_2H_4, with a 1:1 mixture of CO and H_2 is shown in equation 1.

equation 1 $C_2H_4(g) + CO(g) + H_2(g) \rightleftharpoons CH_3CH_2CHO(g)$

At atmospheric pressure a cobalt-based catalyst is used in this reaction.

(i) State and explain the effect of using a catalyst on this reaction.

.....

.....

.....

..... [2]

(iii) Use the information in Table 3.2 to calculate the enthalpy change, ΔH_r, of the reaction in equation 1.

equation 1 $C_2H_4(g) + CO(g) + H_2(g) \rightleftharpoons CH_3CH_2CHO(g)$

Table 3.2

compound	enthalpy change of formation, ΔH_f / kJ mol^{-1}
$C_2H_4(g)$	+52
$CO(g)$	–111
$CH_3CH_2CHO(g)$	–187

ΔH_r = ... kJ mol^{-1} [2]